

Retail Business Performance & Profitability Analysis

1. Introduction

This project analyzes retail transaction data to understand sales performance, profitability, product performance, regional trends, seasonal behavior, and the impact of discounts. The analysis helps identify profitable areas and products that may negatively affect business performance.

2. Abstract

The Superstore retail dataset was analyzed using SQL, Python (Pandas), and Tableau. Data was cleaned and prepared before performing business analysis. SQL was used to calculate sales, profit, profit margins, category and sub-category performance, product profitability, regional performance, seasonal trends, and discount impact. Python was used for exploratory data analysis and visualization. A Tableau dashboard was created to present important business KPIs and trends interactively.

3. Tools Used

- **SQL / MySQL** — Data storage and business analysis
- **Python / Pandas** — Data cleaning and exploratory analysis
- **Matplotlib / Seaborn** — Data visualization
- **Tableau** — Interactive dashboard
- **GitHub** — Project version control and documentation

4. Steps Involved

1. Selected and loaded the Superstore retail dataset.
2. Cleaned and prepared the data using Python and Pandas.
3. Created a MySQL database and imported the cleaned dataset.
4. Performed SQL analysis on sales, profit, products, categories, regions, seasons, and discounts.
5. Performed exploratory analysis using Python.
6. Created an interactive Tableau dashboard containing KPIs and business performance charts.
7. Identified important profitability patterns and business opportunities.

5. Key Findings

- Category and sub-category analysis identifies the strongest and weakest areas of profitability.
- Product-level analysis highlights highly profitable products as well as products generating negative profit.
- Regional analysis shows differences in sales and profitability across regions.
- Monthly and seasonal analysis helps identify periods of stronger and weaker business performance.

- Discount analysis helps evaluate the relationship between discount levels and profitability.
- Slow-moving product candidates can be identified using sales quantity and order frequency.

6. Conclusion

The project demonstrates how SQL, Python, and Tableau can be combined to transform retail transaction data into useful business insights. The analysis can help management improve product strategy, control excessive discounting, focus on profitable categories and regions, and review slow-moving or loss-making products. The Tableau dashboard provides an interactive way for stakeholders to monitor retail performance and support data-driven decisions.